

Ryan Thomsen

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EDUCATION

M.Eng. in Biomedical Engineering; University of Florida, Gainesville, FL Expected 8/2026
B.S. in Biology; minor in Biomedical Physics; University of South Florida, Tampa, FL 5/2025

RELEVANT PROJECTS

Development of Visual and Auditory Stimulation Device to Upregulate Glymphatic System

- Developed a wearable neurostimulation prototype delivering 40 Hz infrared and auditory stimuli
- Utilized embedded programming (Python, C) to test and iterate multiple prototype versions
- Built and modified custom circuitry, including soldering LEDs, op-amps, switches, and wiring before modeling the device in AutoCAD and prepared files for 3D printing

Mechanical Lift for Mobility Impaired Patients to be Placed onto a Healthcare Treatment Table

- Collaborated with a radiation oncologist (Kathryn E. Hitchcock, MD, PhD) to design and develop a mechanical lift for transferring mobility-impaired patients from wheelchairs to a LINAC treatment table
- Combined the use of CAD (OnShape), woodworking, sewing, and power tools to create a proof of concept
- Drafted an SBIR Proposal, learned early-stage FDA regulatory planning (device classification, 510(k))

Commercialization of a Surgical Finger Dissection Device

- Worked with surgeons to develop a bench to bedside approach for an IP protected surgical device. Developed core goals including identifying customers, how the device should be used, and what market plan would best fit the device
- Completed training with the [U.S. National Science Foundation Innovation Corps \(I-Corps™\)](#) program that facilitates the transformation of NSF-funded research projects that are ready to move toward commercialization. The program prepares scientists and engineers to extend their focus beyond the university laboratory

WORK EXPERIENCE

Graduate Research Assistant, [Human Neuromechanics Lab](#), University of Florida ([Dr. Dan Ferris](#)) 8/2025- Present

- Lead and mentor a team of undergraduate students for investigating neural responses to stimuli using EEG systems. Process and analyze EEG data using MATLAB and EEGLAB
- Troubleshoot and optimize Python scripts for stimulus delivery and data synchronization. Recruit and manage human participants, conduct informed consent, ensure IRB compliance, and assist with protocol review and documentation
- Collaborate on studies involving EEG, EMG, OptiTrack motion capture, and VR-based experimental setups
- Develop and maintain lab documentation, authored EEGLAB GUI usage guides, and contributed to GitHub repository for EEG Processing. Assist in reviewing grant proposals, and contribute to efforts securing research funding

Research Intern, [Hyperbaric Oxygen Therapy Center](#), University of South Florida ([Dr. Joseph Dituri](#)) 12/2024- 5/2025

- Supported a clinical trial (NCT06581003) investigating hyperbaric oxygen therapy for traumatic brain injury in veterans and active-duty service members by obtaining and documenting patient blood pressure, pulse, respiratory rate, and blood glucose levels for diabetic participants.
- Operated hyperbaric chambers and monitored patients throughout pressurization and depressurization to ensure safety

LEADERSHIP & INVOLVEMENT

Member, EEG Journal Club 8/2025- Present

- Critically analyze recent peer-reviewed articles that use EEG to understand novel analysis methods and learn of lead researchers in the field

President, USF Motorcycle Club 5/2023- 5/2025

- Petitioned for and obtained over \$10,000 in funding for club events and merchandise. Signed contracts with local vendors for club discounts. Organized and led group rides to various volunteer opportunities for students.

Mentor, Transfer Engagement and Achievement Mentorship (T.E.A.M) 8/2023- 5/2024

- Mentored four students in finding and accessing campus resources, answering questions about campus life and academics, and fostering a growth mindset, leadership skills, and academic success skills

CERTIFICATIONS

	Obtained
Basic Life Saving (BLS)	2/2026
HIPPA Awareness Training	8/2025
IRB Training	12/2025
Chemical Hygiene Plan Training	8/2025
NAUI Advanced Rescue Diver	7/2021
NAUI Diving First Aid	7/2021

SKILLS

CAD (Fusion360, OnShape), Python, MATLAB, C, EEGLAB, EEG, Electrical Circuits, ESP32, Welding/Soldering, 3D Printing, Power Tools

